# Logistic Regression Using Scikit-learn

## Full Source Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

iris = load_iris()
X = iris.data[:100]
y = iris.target[:100]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("Accuracy:", format(accuracy_score(y_test, y_pred), ".2f"))
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("Classification Report:")
print(classification_report(y_test, y_pred))
```

## Full Output

```
Accuracy: 1.00
Confusion Matrix:
[[17  0]
 [ 0 13]]
Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        17
           1       1.00      1.00      1.00        13

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```

**How to run:** Install Scikit-learn with `pip install scikit-learn`. Save the program as `logistic_regression.py` and press F5 in Python IDLE.

**Note:** The program uses the first 100 Iris samples (two classes). The output shown is for the specified train-test split.